

SHIVANI REDDY

AWS Data Engineer | USA | 201-904-0760 | shivanireddyk06@gmail.com | [LINKEDIN](#)

PROFESSIONAL SUMMARY

AWS Data Engineer with 4+ years of experience building secure, scalable data platforms across healthcare and finance.

Hands on experience delivering batch and near real time pipelines on AWS using Amazon S3, AWS Glue, EMR, Lambda, Kinesis, Step Functions, and Airflow for governed analytics and operational reporting.

Strong in SQL, PySpark, API integration, JSON and XML normalization, data quality validation, and secure lakehouse design with IAM, KMS, Lake Formation, and Glue Data Catalog.

Experienced in partnering with stakeholders to translate business and reporting needs into reusable datasets, reliable orchestration, and clear technical documentation.

Improved pipeline reliability, reduced manual triage, and lowered common query and scan overhead through partitioning, replay workflows, observability, and runbook driven operational support.

PROFESSIONAL SKILLS

- **Programming Languages & Scripting:** Python, PySpark, SQL, Bash
- **Data Processing :** Apache Spark on EMR, AWS Glue, batch pipelines, near real time data processing, CDC patterns, schema evolution, data quality validation
- **Data Ingestion & Integration:** Amazon S3, AWS Glue Crawlers, Amazon Kinesis Streams, Amazon Kinesis Firehose, AWS Lambda, REST APIs, JSON normalization, XML normalization, Amazon EventBridge
- **Warehouses & Analytics Engines:** Amazon Redshift, Amazon Athena, relational SQL processing, analytics ready dataset publishing
- **Data Lake & Governance:** Amazon S3 lakehouse, Bronze Silver Gold architecture, Delta Lake, AWS Glue Data Catalog, AWS Lake Formation, partitioning, compaction, source to target mapping
- **Security & Compliance:** IAM, AWS KMS, AWS Secrets Manager, VPC endpoints, PrivateLink, encryption at rest, encryption in transit, HIPAA aligned controls, SOC 2 aligned controls
- **Observability & Reliability:** Amazon CloudWatch, AWS CloudTrail, SLA monitoring, retries, backoff, failure classification, incident triage, runbooks
- **CI/CD & Release Engineering:** GitHub, GitLab, Jenkins, AWS CodeBuild, AWS CodePipeline
- **Infrastructure as Code (IaC):** Terraform, AWS CloudFormation
- **Cost & Performance:** partition pruning, optimized file sizing, incremental loads, deduplication, upserts, compaction, query tuning
- **SDLC & Collaboration:** Agile delivery, Jira, Confluence, stakeholder requirements, documentation, code reviews, mentorship, cross functional collaboration

PROFESSIONAL EXPERIENCE

Lumeris | Cloud Data Engineer | Remote, USA | May 2024 to Present

- Built enterprise ingestion pipelines for claims, policy, and finance data into Amazon S3 using AWS Glue crawlers and catalog standards, reducing schema ambiguity for downstream analytics and improving repeatability of source onboarding.
- Designed PySpark transformations on AWS Glue and EMR to cleanse, standardize, and enrich multi source healthcare data, choosing idempotent processing patterns to protect daily SLAs when late arriving files required reruns.
- Developed curated Bronze, Silver, and Gold datasets with partitioning and compaction strategies, improving Athena and Redshift query performance and lowering unnecessary scan costs for reporting consumers.
- Implemented near real time ingestion with Amazon Kinesis, Lambda, and replay workflows for event validation and routing, reducing manual triage and improving reliability of operational data feeds.
- Strengthened security and governance with AWS Glue Data Catalog, Lake Formation, IAM, KMS, and private network access controls, supporting HIPAA aligned access, encryption, and auditability across shared datasets.
- Orchestrated batch and event driven workflows with Step Functions and Airflow, adding retries, backoff, and failure classification to improve resilience and reduce recurring production incidents.
- Instrumented pipelines with CloudWatch metrics, logs, alarms, and runbooks, accelerating root cause analysis and improving operational response for business critical data products.
- Partnered with analytics and business teams to translate reporting requirements into reusable curated datasets and documented data flows, improving handoff quality and consistency of downstream metric usage.

Algoworks | ETL Developer | Hyderabad, India | Jan 2021 to Jul 2023

- Engineered Spark based ETL pipelines on EMR for financial and regulatory data, producing reconciled outputs required for compliance and month end reporting under tight delivery timelines.
- Built end to end data flows to land raw files on Amazon S3, transform with AWS Glue, and publish analytics ready datasets to Redshift and Athena, improving reliability of scheduled reporting and downstream consumption.
- Developed SQL and PySpark validation rules for schema, completeness, and referential integrity, routing bad records to quarantine paths to reduce downstream reprocessing and recurring data quality issues.
- Implemented incremental load patterns using watermarks, deduplication, upserts, and partition design, reducing reprocessing effort and controlling compute and storage costs during backfills.
- Packaged transformation utilities with Docker and deployed batch utilities on Kubernetes, improving portability and consistency across development, test, and production environments.
- Added CloudWatch monitoring, alerting, and runbook based support procedures, increasing operational visibility and speeding issue resolution during peak reporting cycles.
- Collaborated with reporting stakeholders to define metric logic and delivery expectations, improving alignment between source data, transform rules, and analytics outputs.

Algoworks | Data Engineer Intern | Hyderabad, India | Jun 2020 to Jan 2021

- Wrote SQL based validation checks across source feeds and warehouse staging layers to detect accuracy and completeness issues before scheduled loads reached downstream reports.
- Developed Python ingestion scripts for external REST APIs, normalized JSON and XML payloads into SQL ready structures, and staged curated extracts on Amazon S3 for downstream Spark processing.
- Assisted in building AWS Glue jobs for foundational transformations and curated outputs, documenting business rules and source to target mappings to reduce repeated clarification and onboarding time.
- Created data flow documentation, schema dictionaries, and runbooks to improve operational readiness, audit support, and handoff quality across delivery teams.
- Supported reporting teams with curated extracts and metric definitions, helping standardize business logic used in recurring dashboards and ad hoc data requests.

PROJECTS

- **Enterprise API Data Integration Platform** | Data Engineer | Python, Amazon Kinesis, AWS Lambda, AWS Glue, EMR, Amazon S3, Airflow | Ingested API and webhook sensor data, normalized nested JSON into curated SQL ready datasets, orchestrated replay handling and validation rules, partitioned data by event date and source, monitored near real time SLAs with CloudWatch, improved pipeline reliability by 35%.
- **AWS Lakehouse for Retail Analytics** | Data Engineer | Amazon S3, AWS Glue PySpark, EMR, Step Functions, Athena, Redshift | Designed batch ingestion for 10M+ retail records, built Bronze Silver Gold datasets, handled late arriving files and duplicate loads, added observability and cost controls through partitioning and optimized file sizing, reduced common query latency for reporting use cases.

EDUCATION

- Master of Science, Information Systems and Technology, Wilmington University | 2025
- Bachelor of Technology, Information Technology, Jawaharlal Nehru Technological University | 2022

CERTIFICATIONS

- AWS Certified Solutions Architect Associate
- Databricks Certified Data Engineer Professional
- Microsoft Certified Azure Fundamentals AZ 900
- Verizon Cloud Platform Job Simulation Forage